like multiple strikes of lightning—larger numbers of virtual string pairs are increasingly unlikely to erupt momentarily into existence. If the coupling constant is 1 or greater, however, it is increasingly likely that ever-larger numbers of such virtual pairs will momentarily burst on the scene.5 The upshot is that if the string coupling constant is less than 1, the loop diagram contributions become ever smaller as the number of loops grows. This is just what is needed for the perturbative framework, since it indicates that we will get reasonably accurate results even if we ignore all processes except for those with just a few loops. But if the string coupling constant is not less than 1, the loop diagram contributions become more important as the number of loops increases. As in the case of a trinary star system, this invalidates a perturbative approach. The supposed ballpark approximation—the process with no loops—is not in the ballpark. (This discussion applies equally well to each of the five string theories—with the value of the string coupling constant in any given theory determining the efficacy of the perturbative approximation scheme.)

This realization leads us to the next crucial question: What is the value of the string coupling constant (or, more precisely, what are the values of the string coupling constants in each of the five string theories)? At present, no one has been able to answer this question. It is one of the most important unresolved issues in string theory We can be sure that conclusions based on a perturbative framework are justified only if the string coupling constant is less than 1. Moreover, the precise value of the string coupling constant has a direct impact on the masses and charges carried by the various string vibrational patterns. Thus, we see that much physics hinges on the value of the string coupling constant. And so, let’s take a closer look at why the important question of its value—in any of the five string theories—remains unanswered.

The Equations of String Theory

The perturbative approach for determining how strings interact with one another can also be used to determine the fundamental equations of string theory. In essence, the equations of string theory determine how strings interact and, conversely, the way strings interact directly determine the equations of the theory.